

## UNIVERSITY NAME

*Rigid Body Mechanics — Final Exam**Year: 2014***Exercise 1:**

Let  $R_0(O, \bar{x}_0, \bar{y}_0, \bar{z}_0)$  be a Galilean orthonormal direct frame relative to which we propose to study the motion of a system  $(\Sigma)$  consisting of:

- A straight rod (T) along the direction class  $(O, \bar{x})$ , with negligible cross-section and mass, which can rotate freely around its fixed endpoint O. - A solid homogeneous sphere (S) of mass  $M$ , radius  $a$ , and center G.

The sphere (S) is pierced along its axis  $(G, \bar{x})$  by the rod (T). It can slide along the rod and rotate around it.

All friction is neglected (i.e., the different contacts are considered frictionless) and we assume the connection at O is perfect. The motion parameters of  $(\Sigma)$  in the frame  $R_0$  are the three usual Euler angles  $(\psi, \theta, \phi)$  and  $\overline{OG} = \rho(t)$ .

All results will be expressed in the basis  $(i, \bar{w}, \bar{z})$ .

1. Determine the following instantaneous rotation vectors:

$$\hat{\Omega}(T/R_0), \quad \hat{\Omega}(S/T), \quad \hat{\Omega}(S/R_0)$$

2. Calculate the velocity and acceleration vectors of point G relative to  $R_0$ .
3. Calculate the inertia matrix at point O of the system  $\Sigma$ :

$$\Pi_0(\Sigma)$$

4. Calculate the angular momentum about O of the system  $\Sigma$  relative to  $R_0$ :

$$\tilde{L}_0(\Sigma/R_0)$$

5. Show that:

$$\tilde{L}_0(\Sigma/R_0) \cdot \tilde{z}_0 = \text{const} = k_1 \quad \text{and} \quad \tilde{L}_0(\Sigma/R_0) \cdot \tilde{z} = \text{const} = k_2.$$

Deduce two equations of motion in the form of first integrals.

6. Determine the kinetic energy of the system  $\Sigma$  relative to the frame  $R_0$ .
7. By applying the kinetic energy theorem to  $\Sigma$ , give a third equation of motion in the form of a first integral. (Assume the rod-sphere connection is perfect.)

8. In the special case where the axis  $(O, \bar{z})$  remains coincident with the axis  $(O, \bar{z}_0)$ , determine  $\rho(t)$ . Given:

$$\rho(0) = 0 \quad \text{and} \quad \dot{\rho}(0) = \rho_0$$

Show that:

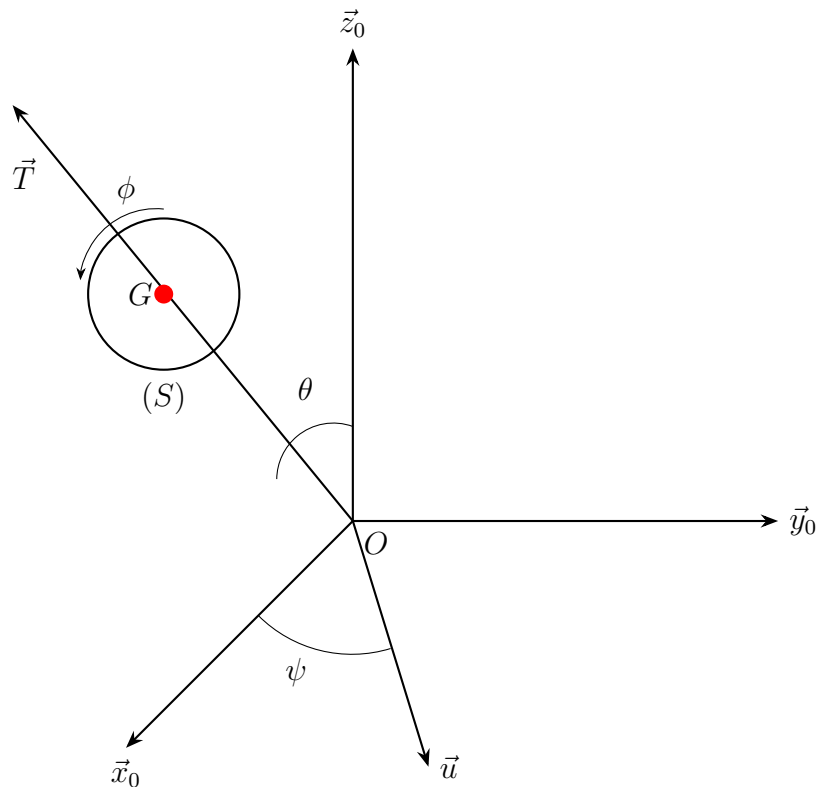
$$\tilde{L}_0(\Sigma/R_0) \cdot \tilde{z}_0 = \text{const} = k_1 \quad \text{and} \quad \tilde{L}_0(\Sigma/R_0) \cdot \tilde{z} = \text{const} = k_2.$$

Deduce two equations of motion in the form of first integrals.

9. Determine the kinetic energy of the system  $\Sigma$  relative to the frame  $R_0$ .
10. By applying the kinetic energy theorem to  $\Sigma$ , give a third equation of motion in the form of a first integral. (Assume the rod-sphere connection is perfect.)
11. In the special case where the axis  $(O, \bar{z})$  remains coincident with the axis  $(O, \bar{z}_0)$ , determine  $\rho(t)$ . Given:

$$\rho(0) = 0 \quad \text{and} \quad \dot{\rho}(0) = \rho_0$$

## System Diagram



**Answer Area**

1. Rotation vectors:

$$\begin{aligned}
\hat{\Omega}(T/R_0) &= \dot{\psi} \bar{z}_0 + \dot{\theta} \bar{u} \\
\hat{\Omega}(S/T) &= \dot{\phi} \bar{x} \\
\hat{\Omega}(S/R_0) &= \hat{\Omega}(T/R_0) + \hat{\Omega}(S/T) \\
&= \dot{\psi} \bar{z}_0 + \dot{\theta} \bar{u} + \dot{\phi} \bar{x}
\end{aligned}$$

where  $\bar{u} = \bar{z}_0 \times \bar{x} / \|\bar{z}_0 \times \bar{x}\|$  in the nodal line direction.

2. Velocity and acceleration of G:

$$\begin{aligned}
\bar{V}(G/R_0) &= \dot{\rho} \bar{x} + \rho \hat{\Omega}(T/R_0) \times \bar{x} \\
&= \dot{\rho} \bar{x} + \rho \dot{\theta} \bar{u} \times \bar{x} + \rho \dot{\psi} \sin \theta \bar{y} \\
\bar{a}(G/R_0) &= \left. \frac{d}{dt} \bar{V}(G/R_0) \right|_{R_0}
\end{aligned}$$

(Full expansion omitted for brevity, follows from differentiation of velocity.)

3. Inertia matrix at O of
- $\Sigma$
- :

Rod (T) mass negligible, only sphere contributes.

Parallel axis theorem:

$$\Pi_O(\Sigma) = \Pi_G(S) + M \begin{pmatrix} \rho^2 & 0 & 0 \\ 0 & \rho^2 + \frac{2}{5}a^2 & 0 \\ 0 & 0 & \rho^2 + \frac{2}{5}a^2 \end{pmatrix}$$

in basis  $(\bar{x}, \bar{w}, \bar{z})$  with  $\bar{w}$  perpendicular to  $\bar{x}$  in the plane of  $\bar{x}$  and  $\bar{z}_0$ .

4. Angular momentum about O:

$$\begin{aligned}
\tilde{L}_O(\Sigma/R_0) &= \tilde{L}_O(S/R_0) \\
&= \mathbf{I}_O(S) \cdot \hat{\Omega}(S/R_0) + M \overline{OG} \times \bar{V}(G/R_0)
\end{aligned}$$

where  $\mathbf{I}_O(S)$  is the inertia tensor of sphere relative to O.

5. Conservation laws:

No external torque about  $\bar{z}_0$  and  $\bar{z}$  axes  $\Rightarrow$

$$\tilde{L}_O(\Sigma/R_0) \cdot \bar{z}_0 = k_1, \quad \tilde{L}_O(\Sigma/R_0) \cdot \bar{z} = k_2$$

These give two first integrals of motion (conservation of angular momentum components along  $\bar{z}_0$  and  $\bar{z}$ ).

6. Kinetic energy:

$$T = \frac{1}{2} M \bar{V}^2(G/R_0) + \frac{1}{2} \hat{\Omega}(S/R_0) \cdot \mathbf{I}_G(S) \cdot \hat{\Omega}(S/R_0)$$

with  $\mathbf{I}_G(S) = \frac{2}{5} M a^2 \mathbf{1}$  (isotropic).

## 7. Kinetic energy theorem:

No non-conservative work (perfect connections, no friction)  $\Rightarrow$

$$\frac{dT}{dt} = 0 \quad \text{or} \quad T = \text{const} = E_0$$

which is a third first integral (conservation of total mechanical energy if no potential).

8. Special case  $\bar{z} = \bar{z}_0$ :

Then  $\theta = 0$ ,  $\bar{u}$  undefined, rotation simplifies:

$$\hat{\Omega}(S/R_0) = \dot{\psi}\bar{z}_0 + \dot{\phi}\bar{x}$$

Conservation laws give:

$$\frac{2}{5}Ma^2(\dot{\psi} + \dot{\phi}) + M\rho^2\dot{\psi} = k_1$$

$$\frac{2}{5}Ma^2\dot{\phi} = k_2$$

Energy conservation:

$$\frac{1}{2}M\dot{\rho}^2 + \frac{1}{2} \cdot \frac{2}{5}Ma^2(\dot{\psi}^2 + \dot{\phi}^2 + 2\dot{\psi}\dot{\phi}) + \frac{1}{2}M\rho^2\dot{\psi}^2 = E_0$$

System reduces to ODE for  $\rho(t)$  using constants  $k_1, k_2, E_0$ .

11. Determination of  $\rho(t)$  in special case:

From energy and angular momentum conservation, after eliminating  $\dot{\psi}, \dot{\phi}$ , equation for  $\rho$  is:

$$\frac{1}{2}M\dot{\rho}^2 + \frac{k_2^2}{2 \cdot \frac{2}{5}Ma^2} + \frac{(k_1 - k_2)^2}{2M\rho^2} = E_0$$

This resembles a 1D central motion:

$$\frac{1}{2}M\dot{\rho}^2 + V_{\text{eff}}(\rho) = E_0$$

with effective potential

$$V_{\text{eff}}(\rho) = \frac{A}{\rho^2} + B, \quad A, B \text{ constants.}$$

Solving with  $\rho(0) = 0, \dot{\rho}(0) = \rho_0$  yields oscillatory or monotonic motion depending on constants.